MIL-I-22767A(AS)
15 October 1967
Superseding
MIL-I-22767(WEPS)
20 February 1961

MILITARY SPECIFICATION

INTERCOMMUNICATION STATIONS

LS-459(*)/AIC and LS-460(*)/AIC

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide efficient intercommunications and receiver audio output combining and isolating functions. The stations shall meet the requirements for use in high performance aircraft where space, weight, and input power are at a premium.

*1.2 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

Earphone	H87-B/U
Microphone Amplifier	AM-3597A/A
Microphone Amplifier	AM-4326/A
Microphone Amplifier	MX-7221/A
Microphone Amplifier	MI-23020

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-6781	Control Panel, Aircraft, Equipment, Rack or Console mounted.
MIL-P- 7788	Plastic, Plate, Lighting
MIL-I-18309	Integrated Electronic Central AN/ASQ-19

FSC 5831

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SPECIFICATIONS

Military

MIL-I-81036	Integrated Electronic Central AN/ASQ-88
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-STD-810	Military Standard, Environmental Test Methods for Aerospace and Ground Equipment.

*2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120

3. REQUIREMENTS

*3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, parts and materials shall be in accordance with Specification MIL-I-18309 and MIL-I-81036.

3.2 Design and Construction

*3.2.1 General - This equipment shall fulfill all applicable requirements of Specifications MIL-I-18309 and MIL-I-81036 for design, construction and workmanship except in the paragraphs as modified below. The equipment described herein shall be of the console type.

*3.2.2 Total Weight - The total weight of the equipment shall be a minimum consistent with good design and shall not exceed 3.0 pounds for the LS-459(*)/AIC and 3.5 pounds for the LS-460(*)/AIC.

*3.2.3 Operational Stability - The equipment shall operate with optimum performance for 1200 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

*3.2.4 Cables and Connectors - External connectors for the equipment shall be as specified under detailed requirements.

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3.2.5 Control Panels - Both intercommunication stations shall mount in a console arrangement when used in aircraft. These units shall meet the requirements of Specification MIL-C-6781, and they shall have a plastic lighting plate meeting the requirements of Specification MIL-P-7788. The lamps shall be connected in parallel with one side grounded to the chassis and the other side being connected to a separate contact, in the electrical connector receptacle for connection to an external +28 volt input lead. The brilliance of the lamps will be controlled externally and the brilliance control shall not be part of the control. The arrangement of controls and all lettering and markings proposed for use on the front panel of the control shall be approved by the procuring activity prior to preproduction testing. Lighting intensity shall be .4 ± .2 foot lamberts.

3.2.6 Mounting Fasteners

3.2.6.1 Intercommunication Station LS-460(*)/AIC - This station shall be equipped with fasteners meeting the requirements of Specification MIL-F-25173 for mounting.

3.2.6.2 Intercommunication Station LS-459(*)/AIC - This station shall be equipped with four 8-32 captive screws for mounting purposes. The dimension between screw centers parallel to the mounting rails shall be 1.125 inches. The dimension between screw centers between mounting rails shall be 5.468 inches.

3.2.7 Service Conditions - In addition to those requirements specified in Specification MIL-I-18309 and MIL-I-81036 the following requirement shall apply.

*3.2.7.1 Acoustical Noise - No malfunction of the equipment shall occur when subjected to the noise field specified by MIL-STD-810 for grade A equipments.

*3.2.8 Power Requirements - The equipment shall give specified performance with the power sources listed in Specification MIL-I-18309 and MIL-I-81036. The power required shall not exceed the following:

(1) D-C Power 28-Volt System

Power (System Total)	20 Watts (less Panel Lights)
Panel Light Power (Per Panel)	5 Watts

Above power provides for operation of two LS-460(*)/AIC units or an LS-460(*)/AIC and an LS-459(*)/AIC operated in a system.

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3.3 Detail Requirements

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

*3.3.2 Frequency Characteristics - The output response of the amplifier when working into a 600-ohm resistive load or 8-ohm resistive load on the appropriate headset connection, shall not exceed ±3 dB relative to the output at 1000 Hz over the frequency range shown in Table 1.

Input Designator	Frequency Range
Microphone	300-2000 Hz
Radio 1-8	300-2000 Hz
Radio 9-20	150-15,000 Hz

TABLE 1 OUTPUT FREQUENCY RANGE

In addition, the microphone response shall be down at least 15 dB at 3000 Hz.

*3.3.3 Power Output - The amplifier shall be internally adjustable to provide a minimum range of 100 to 300 milliwatts of power at 1000 Hz into a 600-ohm or 8-ohm resistive load without producing distortion in excess of the value given in 3.3.4. During service conditions the power output shall not exceed ±3 dB from the power output obtained at standard conditions. Power output shall not be more than 3 dB below the above specifieations when emergency switching is used.

*3.3.4 Distortion - The total harmonic distortion of the amplifier shall not exceed 5 percent for power outputs between 50 and 150 mw and shall not exceed 10% for power outputs between 150 and 300 mw with maximum specified level provided at the radio inptus.

*3.3.5 Intercommunication Volume Control Range - The volume control on either station shall be capable of adjusting the audio output level due to intercommunication from 300 milliwatts to a value at least 37 dB below 300 milliwatts.

*3.3.6 Input Circuits - Receiver and Microphone audio input connections shall be provided at each station in the following manner:

(1) Receiver audio input numbers 1 through 5 and 9 through 17 and the auxiliary audio input of the LS-460(*) shall be subject to attenuation by the Radio Overrride switch at either station according to 3.3.7.

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(2) Receiver audio input numbers 6, 7, 8, 18, 19, and 20 shall not be attenuated by more than that listed in 3.3.7.

(3) A microphone input shall be provided for intercommunication operation or for transmitter modulation.

(4) An auxiliary audio input shall be provided for radio inputs when the LS-460(*) is used in retrofit installation.

*3.3.6.1 Input Levels - The following inputs shall be capable of producing power output at each station according to 3.3.3.

(1) Receiver audio input on numbers 1 through 8 shall require not more than 8.7 V rms.

(2) Receiver audio input on numbers 9 through 20 shall require not more than 4 volts rms.

(3) Microphone input of 0.5 volts rms from a 100-ohm source shall produce the power output as specified.

(4) Auxiliary audio input of 1.11 volts rms applied to pin T of J1 of the LS-460(*) shall produce the power output as specified.

*3.3.6.2 Input Waveform - All input waveforms shall be sinusoidal to obtain power output specified in paragraph 3.3.3.

*3.3.6.3 Input Impedance - The radio inputs shall be isolated from chassis ground with a Hi and Lo external connection available for each input. The input impedance between any corresponding Hi and Lo input shall be 10 K ±1.0 K ohms. The input impedance of the auxiliary audio input in the LS-460(*) shall be 1 K ±100 ohms.

*3.3.6.4 Isolation Between Radio Inputs - Isolation between radio inputs shall be 60 dB or greater when each input Hi is terminated with 600 ohms and when each input Lo is terminated with less than 5 ohms.

*3.3.7 Radio Override - Operation of the Radio Override switch at either station shall attenuate all receiver audio signals on the override inputs at each station by 14 ±2 dB. Audio output due to intercommunication and receiver audio signals at the non-override inputs shall not be attenuated by more than 1.5 dB when the Radio Override switch is operated.

*3.3.8 Headset Output Load Impedance - The intercommunication stations shall operate with a 600-ohm or 8-ohm floating output load impedance applied at the appropriate headset connection.

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*3.3.9 Microphone Power - The LS-459(*) and LS-460(*) shall be capable of providing 12 ±0.5 VDC @ 25 ma on the microphone power line to supply power to an external microphone adapter.

*3.3.10 Microphone/Transmitter Interface - The unit shall provide 0.5 Vrms ±10% into a 100 ohm transmitter modulation load. The transmitter modulator shall provide a minimum of 8 VDC @25 ma DC power for the internal microphone output circuit.

*3.3.11 Common Intercommunications Line - The intercommunications stations shall provide the capability for a common intercommunication line with the other station in the Integrated Electronic Central. The nominal input/output impedance of this line shall be 5 K ohms.

3.3.12 Electrical Connections

*3.3.12.1 Intercommunication Station LS-459(*)/AIC - All connections to this station shall be made through type MS3112E-16-26P for P1, MS3112-16-26PW for P2, and MS3112E-16-26PY for P3 connectors or approved equivalents.

*3.3.12.2 Intercommunication Station LS-460(*)/AIC - All connections to this station shall be made through type MS3112E-22-55P for P2 and MS3112E-18-32PW for P1 connectors or approved equivalents, mounted on the rear.

3.3.13 Controls, Internal - Adjustable resistors located inside the intercommunications stations shall be used as a means of adjusting amplifier gain to balance out production variations.

3.3.14 Controls, Panel Mounted - The following controls shall be located and suitably marked on the front panel.

3.3.14.1 Intercommunication Station LS-460(*)/AIC -

(1) A three-position toggle switch for selecting HOT MIC, COLD MIC, or RADIO OVERRIDE position. The latter position is momentary. This switch is located to the right on the horizontal center line.

(2) A rotary volume control (VOL) for adjusting the amplitude of the intercommunication audio signal delivered to the headset of the operator. This control is located to the left on the horizontal center line.

(3) A three-position rotary switch normally in the normal (NORM) position when all amplifiers are functioning properly and in the back-up (B/U) or emergency (EMER) position when there is an amplifier failure. Which of the latter two positions selected depends upon which amplifier or amplifiers has failed. This switch is located on the vertical center line above the horizontal center line.

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3.3.14.2 Intercommunication Station LS-459(*)/AIC -

(1) A two-position toggle switch for selecting NORMAL or RADIO OVERRIDE position. The latter position is momentary. This switch is located to the right on the horizontal center line.

(2) A rotary volume control (VOL) for adjusting the amplitude of the intercommunication audio signal delivered to the headset of the operator using the unit. This control is located to the left on the horizontal center line.

(3) A three-position rotary swtich normally in the (NORM) position when all amplifiers are functioning properly and in (B/U) or (EMER) position when there is amplifier failure. Which of the latter two positions is selected depends upon which amplifier or amplifiers has failed. This switch is located slightly below the horizontal center line on the vertical center line.

*3.3.15 Form Factor - The dimensions of LS-460(*)/AIC shall not exceed 2-1/4 inches long by 5-3/4 inches wide by 4.20 inches deep, excluding plug connectors. Projection of knobs shall not exceed 1-1/4 inches from the mounting surface. The dimensions of LS-459(*)/AIC shall not exceed 1-1/2 inches long by 5-3/4 inches wide by 5-1/2 inches deep excluding plug connectors. Projection of knobs shall not exceed 1-1/4 inches from the mounting.

4. QUALITY ASSURANCE PROVISIONS

4.1 General- Equipments shall be subjected to the test requirements of specification MIL-I-18309 and MIL-I-81036 to determine compliance with all the applicable requirements.

5. PREPARATION FOR DELIVERY

5.1 General - Preparation for delivery shall be in accordance with Specification MIL-I-18309 and MIL-I-81036.

6. NOTES

*6.1 Use - This equipment is intended to be used as part of the Integrated Electronic Centrals AN/ASQ-107, 19B, 108, and 88B to provide the intercommunication and receiver audio output combining and isolating functions.

6.2 Precedence - When the requirmments of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

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(2) Central Specification - Integrated Electronic Central Specification MIL-I-18309 and Integrated Electronic Central Specification MIL-I-81036 shall have precedence over this specification.

(3) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.10 of Specification MIL-I-18309 and MIL-I-81036. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.4 Revisions - In specification revisions and superseding amendents an asterisk preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-22767A(AS) Intercommunication Station LS-459(*) and LS-460(*)/AIC

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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